CH365 Chemical Engineering Thermodynamics

Lesson 22
Statements of the Second Law and
Derivation of Entropy

BONUS OP

Chemical Engineering Plebe Majors Fair

21 OCT 2025 from ~1220 to ~1350
Thayer Hall Room 336, 368, or 370[1]

30 minutes = 5 points[2]
Max 1.5 hours (15 points)

Notes:
1. We will only occupy one room.
2. Sign in and out on the provided roster with time in and time out. Interact with prospective cadets and answer questions. Stay active. Try not to congregate in “friends” clusters.

Axioms and Postulates

An axiom is an unprovable statement accepted as true because it is self-evident or particularly useful. For example, for any three numbers a, b, and c in a collection of numbers, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$. In other words, the multiplication operation is associative.

An axiom is generally considered to be true but without a clearly defined proof. You just "know that it is true." Nobody can prove that it is correct or disprove that it is incorrect. An axiom is a proposition which is self-evidently true.

A postulate is the same as an axiom but is a statement with higher significance and relates to a specific theory. For example, a postulate of Euclidean geometry is that a straight line can be drawn between two points.

Axioms and postulates are the same and have the same definition.

They differ based on the context they are used or interpreted. The term axiom is used to refer to a statement which is always true in a broad range. A postulate is used in a very limited subject area.

Axiom is an older term while postulate is relatively modern in usage.

Postulates (Axioms) of the 1st Law

Chapter 2, pages 28-31

There exists a form of energy, known as internal energy U, which is an intrinsic property of a system. **Axiom 1, p. 28**

Internal energy is functionally related to the measurable variables that characterize the system (T, V, P, and composition). **Axiom 1, p. 28**

For a closed system, changes in internal energy are given by:

$$\Delta U = Q + W$$

(First Law of Thermodynamics) The total energy of any system and its surrounding is conserved. **Axiom 2, p. 28**

The macroscopic properties of a homogeneous PVT (fluid) system at equilibrium can be expressed as a function of temperature, pressure, molar volume, and composition. **Axiom 3, p. 31**

composition will be important later

Postulates (Axioms) of the 2nd Law

Entropy S is an intrinsic property of any system at equilibrium.

Axiom 4, p. 177

Entropy is functionally related to the measurable state variables that characterize the system (T, V, P, and composition). Axiom 4, p. 177

Differential changes in the total system entropy S^t are given by:

Axiom 4, p. 178

$$dS^t = \frac{dQ_{rev}}{T} \qquad \text{Eq. 5.1}$$

(we will find out why)

(The Second Law of Thermodynamics) The entropy change of any system and its surroundings, considered together, and resulting from any real process, is positive, approaching zero only when the process approaches reversibility.

Axiom 5, p. 178

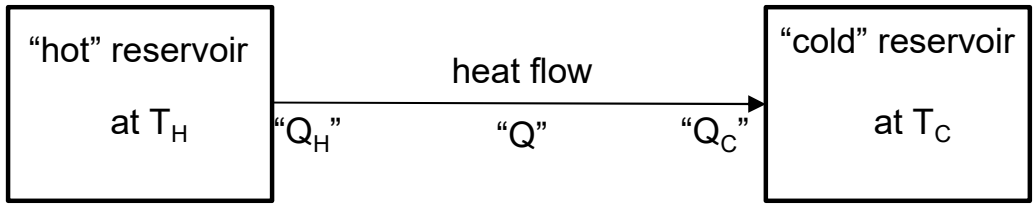
$$\Delta S_{total} \geq 0 \qquad \text{Eq. 5.2}$$

(again, we will find out why)

No process is possible which consists solely in the transfer of heat from one temperature level to a higher one.

Application to Simple Heat Transfer

$$T_H > T_C$$



A "reservoir" is capable of absorbing or rejecting an unlimited quantity of heat without changing temperature.

$$Q_H = -Q_C$$

from Eq. 5.1: $dS^t = \frac{dQ_{rev}}{T} \xrightarrow{\int} \Delta S = \frac{Q}{T}$ (constant T)

$$\Delta S_H^t = \frac{Q_H}{T_H} = \frac{-Q_C}{T_H} \quad \text{and} \quad \Delta S_C^t = \frac{Q_C}{T_C}$$

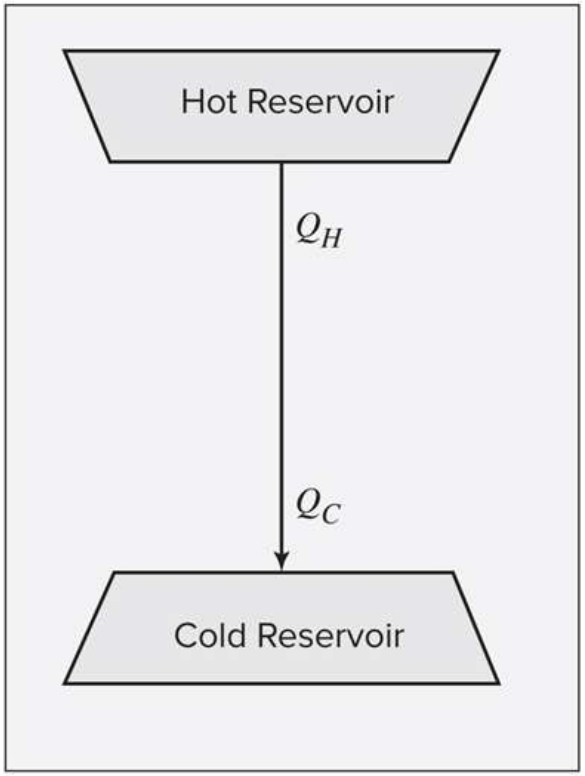
$$\Delta S_{total} = \Delta S_H^t + \Delta S_C^t = \frac{-Q_C}{T_H} + \frac{Q_C}{T_C} = Q_C\left(\frac{T_H - T_C}{T_H \cdot T_C}\right)$$

$$\left.\begin{array}{l} \Delta S_{total} > 0 \\ T_H > 0 \\ T_C > 0 \end{array}\right\} \quad Q_C\left(T_H - T_C\right) > 0 \quad \Rightarrow \quad T_H > T_C \quad Q_C > 0$$

Since Q_C is positive, heat flows into the cold reservoir. That is, **heat flows from higher to lower T**

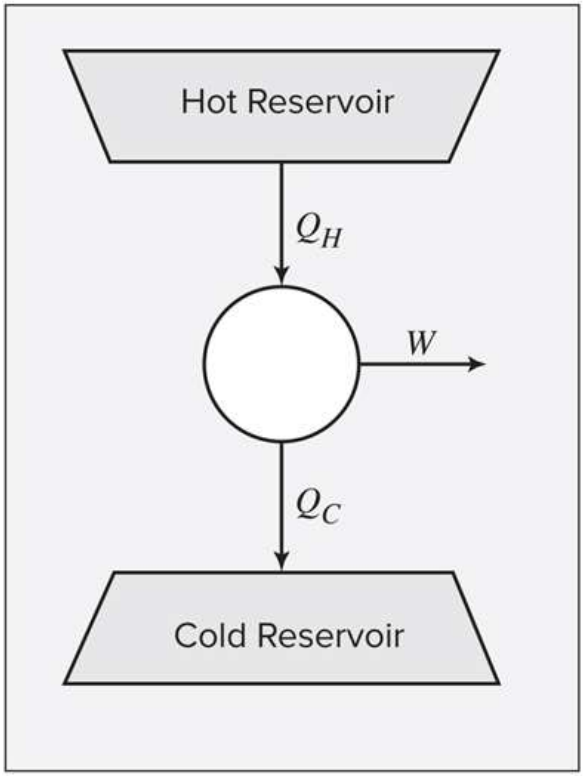
No process is possible which consists solely in the transfer of heat from one temperature level to a higher one. (The process will not run in reverse as is.)

Application to Heat Engines



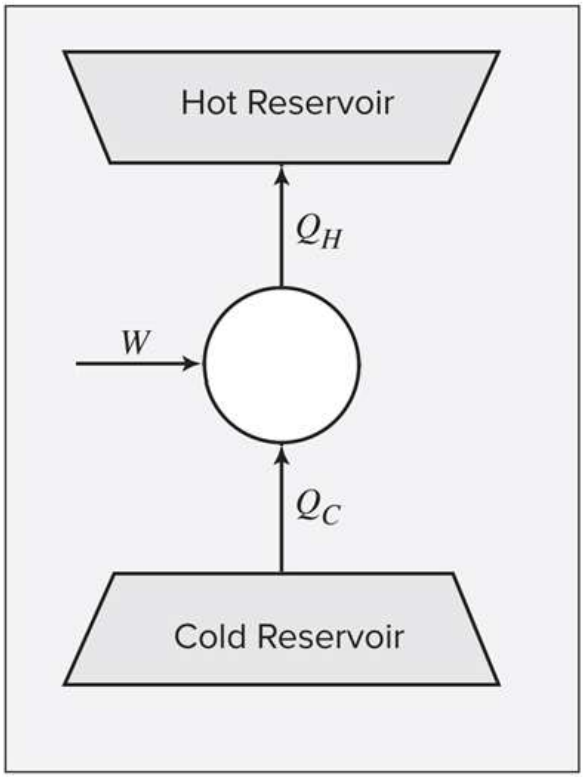
Simple heat transfer

The simple heat transfer model can be modified by adding a device to draw out work from the flowing heat.



Heat engine

Even though no process is possible which consists solely in the transfer of heat from one temperature level to a higher one …



Heat pump

… it turns out that we can do this by adding work to the system.

Derivation of Carnot's Equations

First Law (for the engine):

$$\Delta U = Q + W = Q_H + \overset{<0}{Q_C} + \overset{<0}{W}$$

Cyclic processes (engine):

$$\Delta U = 0$$

$$Q_H = -\underset{<0}{W} - \underset{<0}{Q_C}$$

entropy change for "reversible" engine is 0

Total entropy change (reservoirs):

$$\Delta S_{total} = \frac{\overset{<0}{Q_H}}{T_H} + \frac{Q_C}{T_C}$$

$$Q_H = T_H \Delta S_{total} - \frac{Q_C}{T_C} T_H$$

Equate Q_H and solve for W:

$$-W - Q_C = T_H \Delta S_{total} - \frac{Q_C}{T_C} T_H$$

$$W = -T_H \Delta S_{total} + Q_C \left(\frac{T_H - T_C}{T_C} \right)$$

<u>Limiting case 1:</u>

(engine is completely ineffective)

W=0

$$\Delta S_{total} = Q_C \left(\frac{T_H - T_C}{T_H T_C} \right)$$

Same as simple heat transfer, slide 6

<u>Limiting case 2:</u>

(completely reversible)

ΔS_{total}=0

$$W = Q_C \left(\frac{T_H}{T_C} - 1 \right)$$

Eq. 5.3, p. 181

A "Carnot" engine

Entropy:

equate W from 1st and 2nd Laws

$$-Q_H - Q_C = Q_C \left(\frac{T_H}{T_C} - 1 \right)$$

$Q_C < 0$ wrt engine

$$Q_H = -Q_C \frac{T_H}{T_C}$$

$$\frac{-Q_C}{T_C} = \frac{Q_H}{T_H} \qquad (\Delta S_{total} = 0)$$

Eq. 5.4, p. 182

Efficiency:

$$Q_C = -Q_H \frac{T_C}{T_H}$$

$$W = -Q_H - Q_C$$

$$W = -Q_H + Q_H \frac{T_C}{T_H}$$

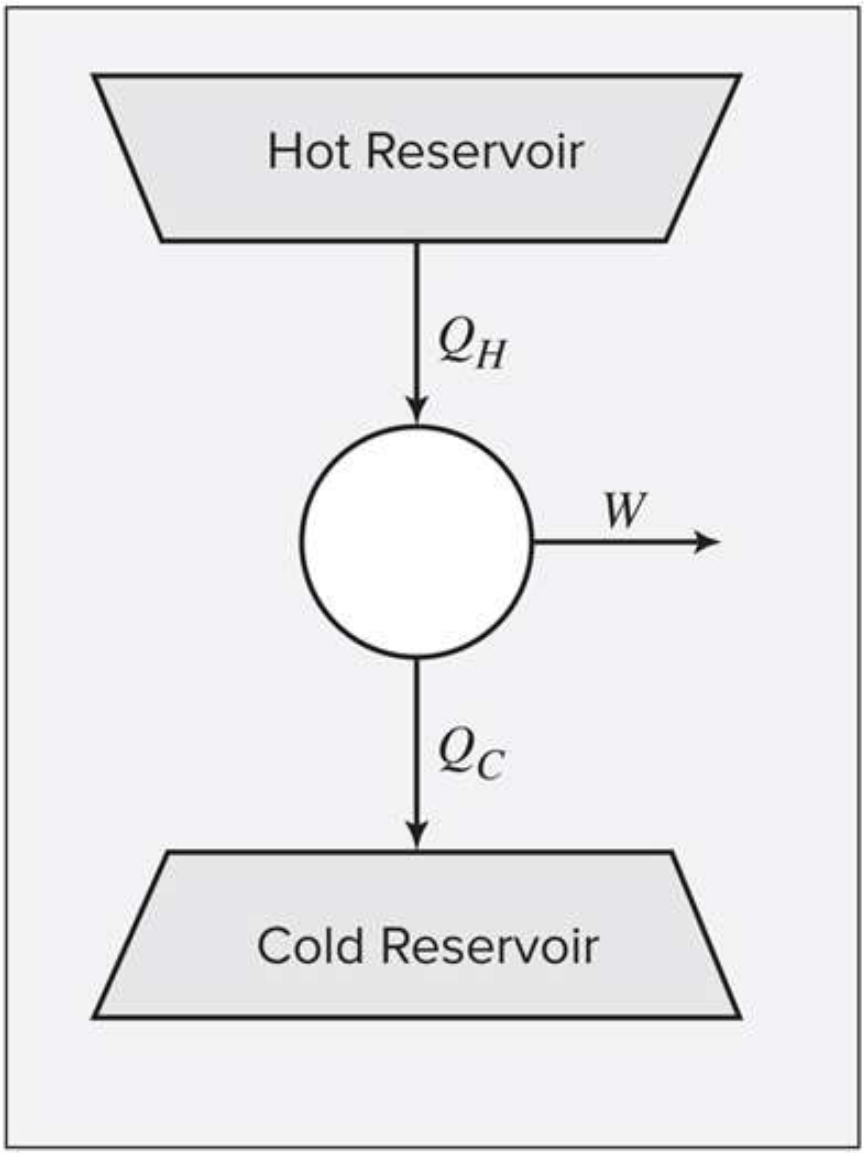
$$W = Q_H \left(\frac{T_C}{T_H} - 1 \right)$$

Eq. 5.5

$$\frac{\overset{<0}{-W}}{Q_H} = 1 - \frac{T_C}{T_H} \equiv \eta_{Carnot}$$

Eq. 5.6, 5.7

Carnot Efficiency



$$\eta \equiv \frac{-W}{Q_H} = 1 - \frac{T_C}{T_H}$$

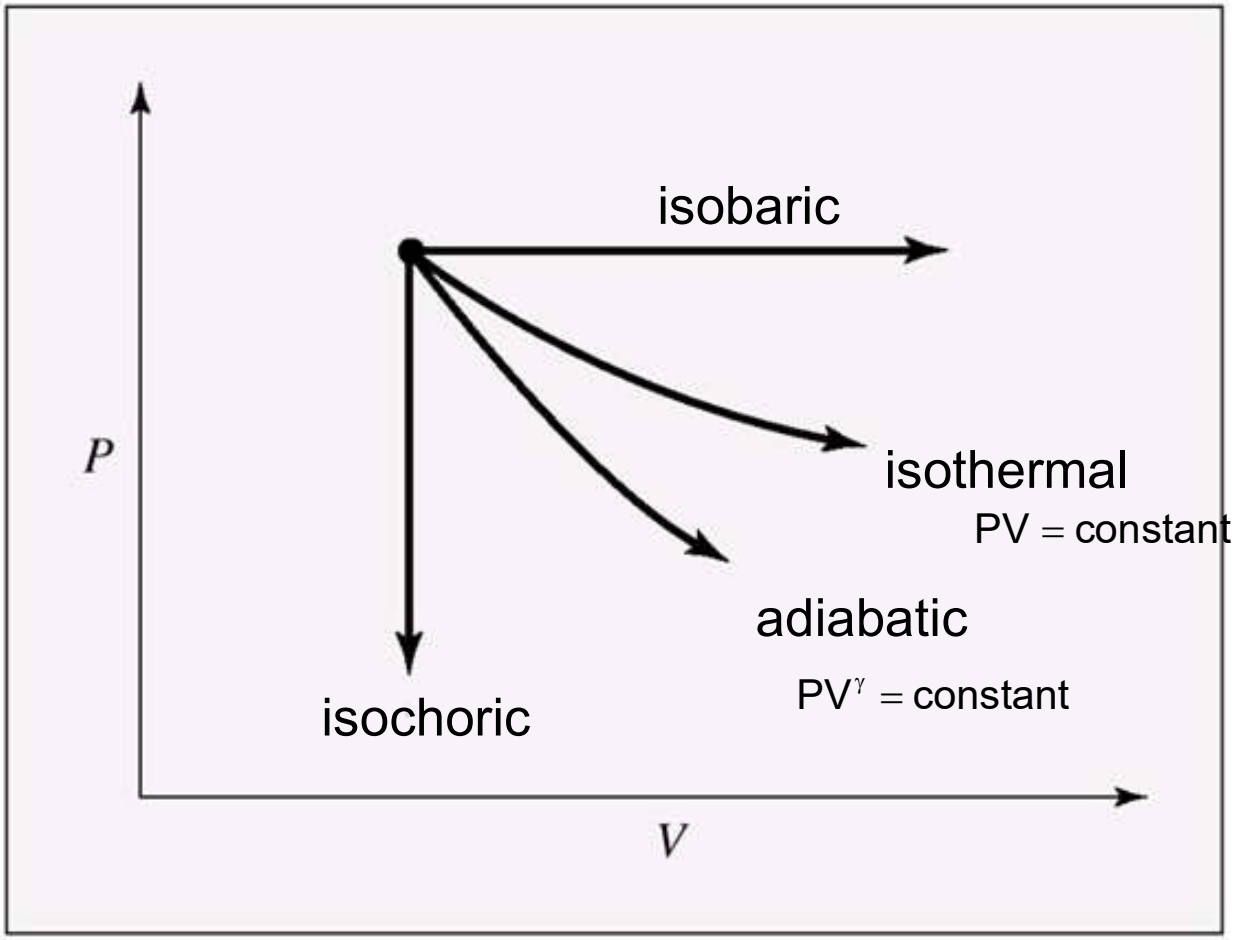
Eqns. 5.6, 5.7

The thermal efficiency of a Carnot engine depends only on the temperature levels and not upon the working substance of the engine.

For two given heat reservoirs, no engine can have a thermal efficiency higher than that of a Carnot engine. The Carnot efficiency is a maximum.

Polytropic Processes – Ideal Gases

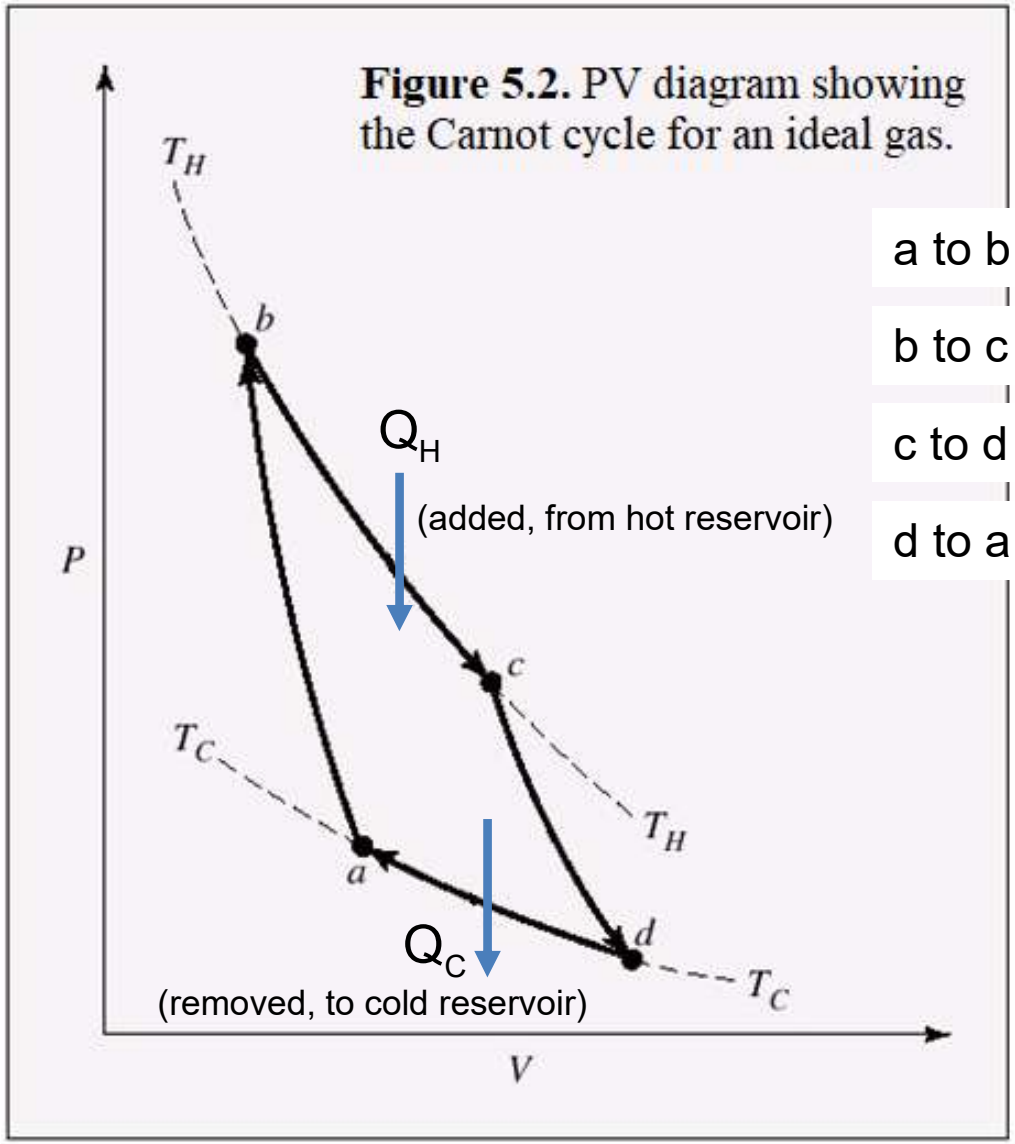
Lesson 11
Slide 13



Paths of polytropic processes

$$\gamma \equiv \frac{C_P}{C_V} > 1$$

Constructing a Carnot Cycle



a to b adiabatic compression T goes from T_C to T_H

b to c isothermal expansion with Q_H added

c to d adiabatic expansion T goes from T_H to T_C

d to a isothermal compression with Q_C ejected

Q, W, ΔH, and ΔU Calculations from L11

$$dQ = C_V dT + RT\frac{dV}{V}$$

(3.16, L11, slide 5, row 2)
(equations for process calcs.)

$$P = \frac{RT}{V} = P(T,V)$$

Lesson 11 Review

Process Calculations, Lesson 11, Slide 6

Example:

Using $P = \frac{RT}{V}$, show that $dW = -\frac{RT}{V}dV$ (3.17) and $dQ = \frac{RT}{V}dV + C_V dT$ (3.16)

$$P = \frac{RT}{V}$$

$$dW = -PdV \quad (1.3) \quad \Rightarrow \quad dW = -\frac{RT}{V}dV \quad \checkmark$$

$$dQ + dW = dU$$

$$dU = C_V dT \quad (3.20a)$$

$$dQ = -dW + C_V dT \quad \Rightarrow \quad dQ = \frac{RT}{V}dV + C_V dT \quad \checkmark$$

(Process calculations for ideal gas)

Derivation of Entropy – Ideal Gases

Isothermal process, equation 3.16, lesson 11, slide 5 , with constant T:

$$Q_H = RT_H \ln\left(\frac{V_c}{V_b}\right) \qquad Q_C = RT_C \ln\left(\frac{V_d}{V_a}\right) \qquad \text{(Eq. 3.20)} \qquad \text{Dividing: } \frac{Q_H}{Q_C} = \frac{RT_H \ln(V_c/V_b)}{RT_C \ln(V_d/V_a)}$$

Adiabatic process, equation 3.16, lesson 11, slide 10, with dQ=0:

$$-\frac{C_V}{R}\frac{dT}{T} = \frac{dV}{V} \qquad \text{(Eq. 3.16)}$$

Integrating 3.16 along the adiabatic paths a-b (T_C-T_H):

$$-\int_{T_C}^{T_H} \frac{C_V}{R}\frac{dT}{T} = \ln\left(\frac{V_b}{V_a}\right) \quad \text{Invert the ratio in the natural log term} \Rightarrow \int_{T_C}^{T_H} \frac{C_V}{R}\frac{dT}{T} = \ln\left(\frac{V_a}{V_b}\right)$$

$$-\int_{T_H}^{T_C} \frac{C_V}{R}\frac{dT}{T} = \ln\left(\frac{V_d}{V_c}\right) \quad \text{Switch the limits of integration} \Rightarrow \int_{T_C}^{T_H} \frac{C_V}{R}\frac{dT}{T} = \ln\left(\frac{V_d}{V_c}\right)$$

These are the same

$$\therefore \ln\left(\frac{V_a}{V_b}\right) = \ln\left(\frac{V_d}{V_c}\right) \Rightarrow \exp\left(\ln\left(\frac{V_a}{V_b}\right)\right) = \exp\left(\ln\left(\frac{V_d}{V_c}\right)\right) \Rightarrow \frac{V_a}{V_b} = \frac{V_d}{V_c} \Rightarrow \frac{V_c}{V_b} = \frac{V_d}{V_a} \Rightarrow \ln\left(\frac{V_c}{V_b}\right) = \ln\left(\frac{V_d}{V_a}\right)$$

$$\frac{Q_H}{Q_C} = \frac{RT_H \ln(V_c/V_b)}{RT_C \ln(V_d/V_a)} \Rightarrow \frac{Q_H}{Q_C} = \frac{T_H}{T_C} \Rightarrow \frac{Q_H}{T_H} + \frac{Q_C}{T_C} = 0$$

"Entropy" Q/T

Recall that we moved around a cyclic path to return to the same state. Therefore, we must conclude Q/T is a state function. If it looks like a duck and quacks like a duck…

Example 5.1

A central power plant, rated at 800,000 kW, generates steam at 585 K and discards heat to a river at 295 K. If the thermal efficiency of the plant is 70% of the maximum possible value, how much heat is discarded to the river at rated power?

Eq. 5.8 $$\eta_{max} \equiv \frac{|W|}{|Q_H|} = 1 - \frac{T_C}{T_H} = 1 - \frac{295}{585} = 0.4957$$

$$\eta = 0.7\eta_{max} = (0.7)(0.4957) = 0.3470$$

$$|W| = 800{,}000\,\text{kW}$$

$$|Q_H| = \frac{|W|}{0.3470} = \frac{800{,}000}{0.3470} = 2{,}305{,}476\,\text{kW}$$

$$|Q_C| = |Q_H| - |W| = 2{,}305{,}476 - 800{,}000 = 1{,}505{,}548\,\text{kW}$$

L22 Equation Summary

$$\Delta U = Q + W = Q_H + Q_L + W$$

$$W = -Q_H - Q_C$$

Eq. 2.3, p. 27

$$\eta \equiv \frac{-W}{Q_H} = \frac{|W|}{|Q_H|} = \text{thermal efficiency}$$

Eq. 5.6, p. 182

$$\eta \equiv \frac{|Q_H| - |Q_C|}{|Q_H|} = 1 - \frac{|Q_C|}{|Q_H|}$$

$$\frac{-Q_C}{T_C} = \frac{Q_H}{T_H} \quad \Rightarrow \quad \frac{|Q_C|}{|Q_H|} = \frac{T_C}{T_H}$$

Eq. 5.4, p. 182

$$\eta = 1 - \frac{T_C}{T_H}$$

Eq. 5.7, p. 182

Questions?